

We focus on the unit tangent vector $\mathbf{T}$, where, as in Section 12.3, we think of $\mathbf{T} = \frac{d\mathbf{r}}{ds}$. This vector changes its direction as a particle goes around a bend, however, its length never changes. The rate at which its direction changes is known as *curvature* and it is the magnitude of $\frac{d^2\mathbf{r}}{ds^2}$:

Definition. If $\mathbf{T}$ is the unit vector of a smooth curve, the **curvature** function of the curve is

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\|$$

Some immediate observations:

- (i) If κ is large for some t_0 -value, then $\mathbf{T}$ is turning sharply at as it passes by the point indicated by $\mathbf{r}(t_0)$. Thus, the curvature of the curve is large at this point.
- (ii) Similarly, if κ is close to zero, then the curvature is small.

Since we typically think of $\mathbf{T}$ as $\mathbf{T}(t) = \frac{\mathbf{v}(t)}{\|\mathbf{v}(t)\|}$, where $\mathbf{v}(t) = \mathbf{r}'(t)$, it can be useful to reinterpret κ in the following way:

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\| = \left\| \frac{d\mathbf{T}}{dt} \frac{dt}{ds} \right\| = \frac{1}{\left| \frac{ds}{dt} \right|} \left\| \frac{d\mathbf{T}}{dt} \right\| = \frac{1}{\|\mathbf{v}\|} \left\| \frac{d\mathbf{T}}{dt} \right\| = \left\| \frac{\mathbf{T}'(t)}{\mathbf{r}'(t)} \right\|$$

This provides an alternate way to think of curvature, where the derivatives are taken *with respect to time*.

It turns out there are simple interpretations of these formulas depending on what setting your curve C lives in:

- If C is a curve in 3-space, then we can calculate κ by

$$\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}.$$

- If C is a curve in the xy -plane, then we can calculate κ by...

- If $y = f(x)$, then $\kappa(x) = \frac{|f'(x)|}{[1 + (f'(x))^2]^{3/2}}$
- If $\mathbf{r}(t) = \langle x(t), y(t) \rangle$, then $\kappa(t) = \frac{|x'(t)y''(t) - x''(t)y'(t)|}{[(x'(t))^2 + (y'(t))^2]^{3/2}}$

Example. Suppose you have a line, L , that passes through the point $P(x_0, y_0, z_0)$ and travels parallel to $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$. Then the points on L are the terminal points on the vector

$$\mathbf{r}(t) = \langle x_0 + v_1 t, y_0 + v_2 t, z_0 + v_3 t \rangle.$$

Show that the curvature of the line L is equal to 0, as expected.

Example. Consider the circle centered at $(0, 0)$ of radius $a > 0$ given by $\mathbf{r}(t) = \langle a \cos(t), a \sin(t) \rangle$. Show that its curvature is constant, as expected.

Example. Calculate the curvature of the curve given by $\mathbf{r}(t) = \langle 2t, 3t^2, 4t^3 \rangle$ at the point $(4, 12, 32)$.

Note that $\mathbf{T} = \frac{d\mathbf{r}}{ds}$ is a *unit* vector that is tangent to the curve and therefore it has constant length. It is this fact that guarantees that its derivative

$$\frac{d\mathbf{T}}{ds}$$

is a vector that is orthogonal to $\mathbf{T}$ (we derived this fact at the end of Section 12.1). Not only is this vector orthogonal to $\mathbf{T}$, but it also *points in the direction the curve is turning!* Thus, if we divide it by its length (which, conveniently, is κ), then we get

$$\mathbf{N}(s) = \frac{1}{\kappa} \frac{d\mathbf{T}}{ds},$$

the **principal unit normal vector**. Equivalently, $\mathbf{N}(t) = \frac{\frac{d\mathbf{T}}{dt}}{\left\|\frac{d\mathbf{T}}{dt}\right\|} = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$ (see footnote ¹)

Example. Consider the standard unit circle traced out by $\mathbf{r}(t) = \langle \cos(t), \sin(t) \rangle$.

(i) Compute $\mathbf{T}(t)$ and $\mathbf{N}(t)$.

(ii) Draw a picture to verify that $\mathbf{N}(t)$ points toward the circle's center.

¹Since $\frac{d\mathbf{T}}{ds} = \frac{d\mathbf{T}}{dt} \frac{dt}{ds} = \frac{d\mathbf{T}}{dt} \frac{1}{\frac{ds}{dt}} = \frac{d\mathbf{T}}{dt} \frac{1}{\|v(t)\|}$, we also have $\mathbf{N} = \frac{\frac{d\mathbf{T}}{ds}}{\left\|\frac{d\mathbf{T}}{ds}\right\|} = \frac{\frac{d\mathbf{T}}{dt} \frac{1}{\|v(t)\|}}{\left\|\frac{d\mathbf{T}}{dt} \frac{1}{\|v(t)\|}\right\|} = \frac{\frac{d\mathbf{T}}{dt}}{\left\|\frac{d\mathbf{T}}{dt}\right\|} = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$.